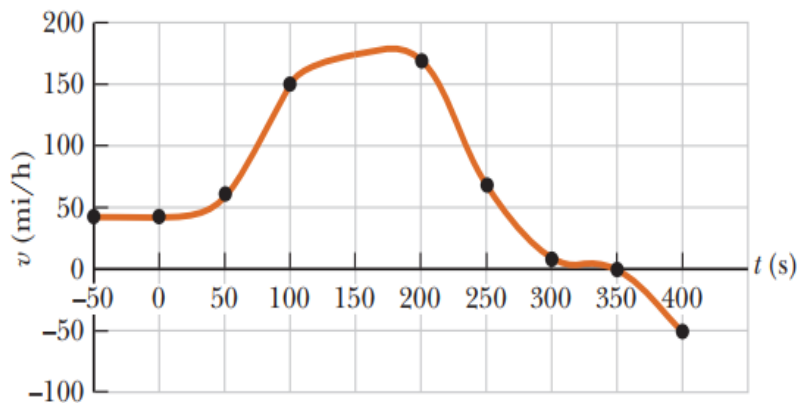


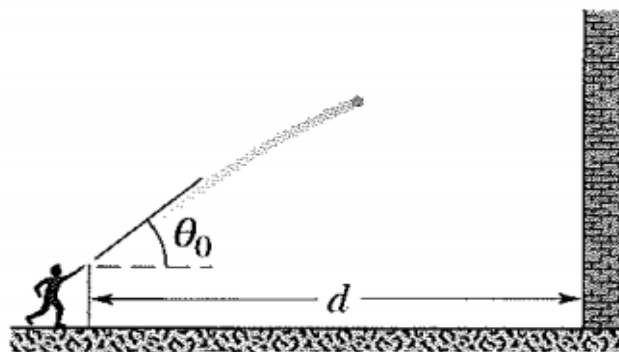
Department of Mathematics and Physics
North south University
Course Title: Physics-I
Course No.: PHY107
Assignment 1
Full marks: 100
Answer All of the Following Questions

1. A velocity vs. time graph for the Acela train is shown in Figure below. (a) Describe the motion of the Acela. (b) Find the peak acceleration of the Acela in miles per hour per second ((mi/h)/s) as the train speeds up from 45 mi/h to 170 mi/h. (c) Find the train's displacement in miles between $t=0$ and $t = 200$ s. (d) Find the average acceleration of the Acela and its displacement in miles in the interval from 200 s to 300 s. (e) Find the total displacement in the interval from 0 to 400 s.



2. A rocket moves straight upward, starting from rest with an acceleration of $+29.4 \text{ m/s}^2$. It runs out of fuel at the end of 4.00 s and continues to coast upward, reaching a maximum height before falling back to Earth. (a) Find the rocket's velocity and position at the end of 4.00 s. (b) Find the maximum height the rocket reaches. (c) Find the velocity the instant before the rocket crashes on the ground.
3. An ice hockey player is standing on his skates on a frictionless field (frozen pond) when an opposing player, moving with a uniform speed of 12 m/s, skates by with the puck. After 3.0 s, the first player makes up his mind to chase his opponent. If he accelerates uniformly at 4.0 m/s^2 , (a) how long does it take him to catch his opponent, and (b) how far has he traveled in that time? (Assume the player with the puck remains in motion at constant speed.)
4. Suppose you are driving a car which can stop in 10 s when traveling at maximum speed. On one occasion, you see a dog a distance d in front of the car and slams on the brakes instantly. The car reaches the dog 8.0 s later, and the dog jumps off the track just in time. If the car travels 4.0 m beyond the position of the dog before coming to a stop, how far was the car from the dog?
5. In a 100-m race, accelerating uniformly, Laura takes 2.00 s and Raisa 3.00 s to attain their maximum speeds, which they each maintain for the rest of the race.

- They cross the finish line simultaneously, both setting a world record of 10.4 s. (a) What is the acceleration of each sprinter? (b) What are their respective maximum speeds? (c) Which sprinter is ahead at the 6.00-s mark, and by how much? (d) What is the maximum distance by which Raisa is behind Laura, and at what time does that occur?
6. Suppose, you released a stone from rest from a certain height, the stone takes 1.50 s to travel the last 30.0 m before it hits the ground. (a) Find the velocity of the stone when it is 30.0 m above the ground. (b) Find the total distance the stone travels during the fall.
 7. A motor car travels 20.0 km due north and then 35.0 km in a direction 60.0° west of north. Using a graph, find the magnitude and the direction of a single vector that gives the net effect of the car's trip.
 8. Suppose you are on travel, you first fly to city A, located 175 km away in a direction 30.0° north of east from your position. Next, you fly for 150 km 20.0° west of north, to city B. Finally, you fly 190 km due west, to city C. (a) Sketch your journey. (b) Find the location of city C relative to the location of your starting point.
 9. Two vectors are presented as $\vec{a} = 3.0\hat{i} + 6.0\hat{j}$ and $\vec{b} = 2.0\hat{i} + 5.0\hat{j}$. Find (a) $\vec{a} \times \vec{b}$, (b) $\vec{a} \cdot \vec{b}$ (c) $(\vec{a} + \vec{b}) \cdot \vec{b}$, and (d) the component of $\vec{a}$ along the direction of $\vec{b}$.
 10. In the product $\vec{F} = q\vec{v} \times \vec{B}$, take $q = 2$, $\vec{v} = 2.0\hat{i} + 4.0\hat{j} + 6.0\hat{k}$ and $\vec{F} = 4.0\hat{i} - 20\hat{j} + 12\hat{k}$. What then is $\vec{B}$ in unit-vector notation and magnitude angle notation if $B_x = B_y$?
 11. A proton initially has $\vec{v} = 4.0\hat{i} - 2.0\hat{j} + 3.0\hat{k}$ and then 4.0 s later has $\vec{v} = -2.0\hat{i} - 2.0\hat{j} + 5.0\hat{k}$ (in meters per second). For that 4.0 s, what are (a) the proton's average acceleration $\vec{a}_{avg}$ in unit vector notation, (b) the magnitude of $\vec{a}_{avg}$, and (c) the angle between $\vec{a}_{avg}$ and the positive direction of the x-axis?
 12. Determine the value of a so that $\vec{A} = 2\hat{i} + a\hat{j} + \hat{k}$ and $\vec{B} = 4\hat{i} - 2\hat{j} - 2\hat{k}$ are perpendicular to each other. What will be the value of a if the two vectors are parallel?
 13. You throw a ball toward a wall at speed 25.0 m/s and at angle $\theta = 40.0^\circ$ above the horizontal (Fig). The wall is distance $d = 22.0$ m from the release point of the ball. (a) How far above the release point does the ball hit the wall? What are the (b) horizontal and (c) vertical components of its velocity as it hits the wall? (d) When it hits, has it passed the highest point on its trajectory?



14. During a tennis match, a player serves the ball at 23.6 m/s, with the center of the ball leaving the racquet horizontally 2.37 m above the court surface. The net is 12 m away and 0.90 m high. When the ball reaches the net, (a) does the ball clear it and (b) what is the distance between the center of the ball and the top of the net? Suppose that, instead, the ball is served as before but now it leaves the racquet at 5.00° below the horizontal. When the ball reaches the net, (c) does the ball clear it and (d) what now is the distance between the center of the ball and the top of the net?
15. Suppose a stone is thrown upward from the level ground in such a way that the maximum height of its flight is equal to its horizontal range R . (a) At what angle θ is the stone thrown? (b) In terms of the original range R , what is the range R_{max} the stone can attain if it is launched at the same speed but at the optimal angle for maximum range? (c) Would your answer to part (a) be different if the stone is thrown with the same speed on a different planet? Explain.
16. An American football player must kick a football from a point 36.0 m from the goal. Half the crowd hopes the ball will clear the crossbar, which is 3.05 m high. When kicked, the ball leaves the ground with a speed of 20.0 m/s at an angle of 53.0° to the horizontal. (a) By how much does the ball clear or fall short of clearing the crossbar? (b) Does the ball approach the crossbar while still rising or while falling?
17. A projectile is launched with an initial speed of 60.0 m/s at an angle of 30.0° above the horizontal. The projectile lands on a hillside 4.00 s later. Neglect air friction. (a) What is the projectile's velocity at the highest point of its trajectory? (b) What is the straight-line distance from where the projectile was launched to where it hits its target?
18. Suppose you ride a carnival Ferris wheel at radius 15 m, completing five turns about its horizontal axis every minute. What are (a) the period of the motion, the (b) magnitude and (c) direction of your centripetal acceleration at the highest point, and the (d) magnitude and (e) direction of your centripetal acceleration at the lowest point?
19. A boy whirls a stone in a horizontal circle of radius 1.5 m and at height 2.0 m above level ground. The string breaks, and the stone flies off horizontally and strikes the

ground after traveling a horizontal distance of 10 m. What is the magnitude of the centripetal acceleration of the stone during the circular motion?

20. A cat rides a merry-go-round turning with uniform circular motion. At time $t_1 = 2.00$ s, the cat's velocity is $\vec{v}_1 = (3.00\text{m/s})\hat{i} + (4.00\text{m/s})\hat{j}$, measured on a horizontal xy coordinate system. At $t_2 = 5.00$ s, the cat's velocity is $\vec{v}_2 = (-3.00\text{m/s})\hat{i} + (-4.00\text{m/s})\hat{j}$. What are (a) the magnitude of the cat's centripetal acceleration and (b) the cat's average acceleration during the time interval $t_2 - t_1$, which is less than one period?